
Satellite thermal vacuum testing (Credit: www.nasa.gov)

vate companies that spend millions or even billions of dollars a year on satellite technology need to ensure that their investments would be fruitful. That's why it's vital to test each satellite properly before it gets launched into space.

Satellite testing approach

Testing must begin at the initial phase of satellite construction for checking each component's acceleration during launch, power consumption, RF performance, etc. Common components that need testing include solar panels, antennas, batteries, electrical equipment, fuel cells, and radio equipment for transmitting information from the satellite to the ground station.

As these parts get assembled into larger pieces, they must undergo additional tests. At the final stage, a quality control team must conduct rigorous testing before declaring the satellite ready for flight.

Requirements for satellite testing depend on its construction, design, and payload. Once satellites are in orbit, these are usually beyond the reach of repair. That is why companies that design and test satellites must consider every contingency and examine everything multiple times to avoid any error. Although, if a part of the satellite fails after launch but other components that are providing critical information are still operational, the satellite will remain in use until it has reached the end of its lifecycle.

A crucial part of all satellite testing is the recording of data that will later get analysed to find and repair problems with the satellite. Test throughput and cost of the test are major contributors to these structures. Regular verification of test equipment performance as per specifications ensures repeatable measurements and successful acceptance testing, which can significantly reduce the time and cost of tests.

Challenges of satellite testing

Satellite testing presents unique challenges. When you test a satellite, you are often testing the one that will eventually go into orbit. Therefore, while the tests need to be meticulous, the testing itself should not damage the satellite in any way.

It costs hundreds of millions of dollars to manufacture and launch a satellite. Currently, satellites are simply decommissioned when these run out of fuel. Satellites are subject to the most challenging technological standards and requirements while operating in space.

Once the satellite is built, no part of it is generally removed for testing. Hence, without being able to see the inside of the satellite, you need to perform all the checks. Thus, sufficiently thorough and accurate preflight testing is required within provided schedules and budgets. The characteristics of satellite communications such as propagation delay, variable channel, and limited bandwidth present challenges for the service providers.

Various tests performed on integrated satellite

Reducing the cost of tests can significantly reduce the total cost of a satellite program, enabling profitable ventures in new satellite applications. Maintaining instrument and system accuracy and regular checking of the key parameters reduce the overall cost of tests and shorten the program schedule in multiple ways. Following are a few major satellite tests:

Thermo-vacuum testing. In this test, the satellite is kept in a vacuum chamber to check the working of its hardware under various thermal loads. The thermal insulation has to maintain the temperatures of the PCBs and sensors like magnetometer and GPS, and also the battery, in the desired range. The functioning of each PCB, sensor, actuator, battery, uplink, and downlink is thoroughly tested within the extreme space temperature range. Thermal vacuum chambers are used to test satellite operating performance in a space environment (vacuum) with great temperature fluctuations to ensure that thermal control analysis performed during the design phase is accurate.

Antenna deployment test. It helps to ensure successful deployment of satellite antenna in space. For establishing a reliable communication between the ground control station and satellite, antenna deployment is one of the most important tasks. It includes antenna design validation in transmitting and receiving mode to test transmitting/receiving modules and verify switch matrix performance to meet the highest performance requirements in the hazards of space.

Global Navigation Satellite System (GNSS) testing. GNSS receiver tests can only be conclusive